ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 2	Alternate digit sums must differ by 0 or 11
2	**(a)** 0	The number must be even and its digit sum a multiple of 3
3	**(c)** 8262	Largest arrangement minus smallest arrangement
4	**(c)** 60	Find the LCM
5	**(b)** 3	4829 leaves remainder 5 when divided by 8
6	**(a)** 1001	Smallest 4-digit number is 1000; add what is needed to reach a multiple of 7
7	**(a)** 45	Subtract the reverse
8	**(c)** 2	Alternate digit sums must differ by 0 or 11
9	**(c)** 0	The number must be even and its digit sum a multiple of 3
10	**(a)** 6084	Largest arrangement minus smallest arrangement

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	8456 divided by 11 leaves remainder 8 (1); add $11 - 8 = 3$ (1).	1+1
12	By 3: Yes, digit sum 15 is a multiple of 3 (1); by 4: Yes, last two digits 16 are divisible by 4 (1).	1+1
13	842-248=594 (1); $594 = 99 \times 6$, so divisible (1).	1+1
14	3721 divided by 8 leaves remainder 1 (1); add $8 - 1 = 7$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LCM of 12 and 15 is 60 (1); first multiple not below 1000 (1); 1020 (1).	1+1+1
16	Alternate sums for each trial; need difference 0 or 11 (1); * $= 0$ (1); check 8305 (1).	1+1+1
17	Middle number $= \frac{54}{3} = 18$ (1); numbers 17, 18, 19 (1); check sum $= 54$ (1).	1+1+1
	OR: LCM of 14 and 21 is 42 (1); first multiple not below 1000 (1); 1008 (1).	1+1+1
18	Alternate sums for each trial; need difference 0 or 11 (1); * $= 9$ (1); check 6919 (1).	1+1+1
	OR: Middle number $= \frac{81}{3} = 27$ (1); numbers 26, 27, 28 (1); check sum $= 81$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Steps: 9541-1459=8082; 8820-0288=8532; 8532-2358=6174 (3); total 3 steps (2).	3+2
	OR: LCM(6, 8, 12) $= 24$ (3); required number $= 24 + 3 = 27$ (2).	3+2
20	Steps: 8620-0268=8352; 8532-2358=6174 (3); total 2 steps (2).	3+2
	OR: LCM(10, 15, 20) $= 60$ (3); required number $= 60 + 4 = 64$ (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 24	Find the LCM
2	**(c)** 8	6713 leaves remainder 3 when divided by 11
3	**(c)** 1008	Smallest 4-digit number is 1000; add what is needed to reach a multiple of 12
4	**(b)** 63	Subtract the reverse
5	**(d)** 4	Alternate digit sums must differ by 0 or 11
6	**(c)** 2	The number must be even and its digit sum a multiple of 3
7	**(c)** 6264	Largest arrangement minus smallest arrangement
8	**(d)** 180	Find the LCM
9	**(d)** 1	5207 leaves remainder 5 when divided by 6
10	**(a)** 1005	Smallest 4-digit number is 1000; add what is needed to reach a multiple of 15

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	By 3: Yes, digit sum 21 is a multiple of 3 (1); by 4: Yes, last two digits 48 are divisible by 4 (1).	1+1
12	753-357=396 (1); $396 = 99 \times 4$, so divisible (1).	1+1
13	6094 divided by 12 leaves remainder 10 (1); add $12 - 10 = 2$ (1).	1+1
14	By 3: Yes, digit sum 18 is a multiple of 3 (1); by 4: No, last two digits 50 are not divisible by 4 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LCM of 16 and 24 is 48 (1); first multiple not below 1000 (1); 1008 (1).	1+1+1
16	Alternate sums for each trial; need difference 0 or 11 (1); * $= 5$ (1); check 8547 (1).	1+1+1
17	Middle number $= \frac{111}{3} = 37$ (1); numbers 36, 37, 38 (1); check sum $= 111$ (1).	1+1+1
	OR: LCM of 18 and 27 is 54 (1); first multiple not below 1000 (1); 1026 (1).	1+1+1
18	Alternate sums for each trial; need difference 0 or 11 (1); * $= 0$ (1); check 5082 (1).	1+1+1
	OR: Middle number $= \frac{150}{3} = 50$ (1); numbers 49, 50, 51 (1); check sum $= 150$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Steps: 8731-1378=7353; 7533-3357=4176; 7641-1467=6174 (3); total 3 steps (2).	3+2
	OR: LCM(9, 12, 18) $= 36$ (3); required number $= 36 + 5 = 41$ (2).	3+2
20	Steps: 9542-2459=7083; 8730-0378=8352; 8532-2358=6174 (3); total 3 steps (2).	3+2
	OR: LCM(8, 12, 20) $= 120$ (3); required number $= 120 + 7 = 127$ (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 27	Subtract the reverse
2	**(b)** 3	Alternate digit sums must differ by 0 or 11
3	**(a)** 1	The number must be even and its digit sum a multiple of 3
4	**(c)** 9081	Largest arrangement minus smallest arrangement
5	**(b)** 70	Find the LCM
6	**(d)** 8	9154 leaves remainder 1 when divided by 9
7	**(a)** 1008	Smallest 4-digit number is 1000; add what is needed to reach a multiple of 18
8	**(d)** 9	Subtract the reverse
9	**(a)** 7	Alternate digit sums must differ by 0 or 11
10	**(d)** 2	The number must be even and its digit sum a multiple of 3

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	961-169=792 (1); $792 = 99 \times 8$, so divisible (1).	1+1
12	5518 divided by 7 leaves remainder 2 (1); add $7 - 2 = 5$ (1).	1+1
13	By 3: Yes, digit sum 15 is a multiple of 3 (1); by 4: Yes, last two digits 24 are divisible by 4 (1).	1+1
14	634-436=198 (1); $198 = 99 \times 2$, so divisible (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	LCM of 20 and 30 is 60 (1); first multiple not below 1000 (1); 1020 (1).	1+1+1
16	Alternate sums for each trial; need difference 0 or 11 (1); * $= 8$ (1); check 3784 (1).	1+1+1
17	Middle number $= \frac{213}{3} = 71$ (1); numbers 70, 71, 72 (1); check sum $= 213$ (1).	1+1+1
	OR: LCM of 9 and 25 is 225 (1); first multiple not below 1000 (1); 1125 (1).	1+1+1
18	Alternate sums for each trial; need difference 0 or 11 (1); * $= 6$ (1); check 9636 (1).	1+1+1
	OR: Middle number $= \frac{333}{3} = 111$ (1); numbers 110, 111, 112 (1); check sum $= 333$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Steps: 7610-0167=7443; 7443-3447=3996; 9963-3699=6264; 6642-2466=4176; 7641-1467=6174 (3); total 5 steps (2).	3+2
	OR: LCM(4, 14, 21) $= 84$ (3); required number $= 84 + 2 = 86$ (2).	3+2
20	Steps: 8621-1268=7353; 7533-3357=4176; 7641-1467=6174 (3); total 3 steps (2).	3+2
	OR: LCM(12, 16, 24) $= 48$ (3); required number $= 48 + 6 = 54$ (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.